

Practical 6

4 March 2026

1. Using AL2CO server, obtain the positional conservation scores from multiple sequence alignment (MSA) of given set of protein sequences (set1 and set2) using the methods given below:

- (i) Unweighted frequency and entropy-based measure
- (ii) Unweighted frequency and variance-based measure
- (iii) Unweighted frequency and sum of pairs measure
- (iv) Weighted frequency and variance-based measure
- (v) Normalize the scores obtained with (i)

Sequences:

Set 1

P69905, P01946, P01942, P01966, P01958, P01959, P01965, P06635, P60529, P80043 and P01980

Set 2

TPIS_HUMAN, TPIS_YEAST, TPIS_GRAGA, TPIS_TRYCR, TPIS_MAIZE, TPIS_MOUSE, TPIS_DROME, TPIS_RABIT, TPIS_CAEEL

Instructions:

A. Access the lab Linux system through SSH (ssh username@ip)

(Ex. *ssh bt3040@10.21.49.164*)

Username: bt3040

Password: bioinfo@123

IP address: 10.21.49.164

Command: al2co

Once you type the command (**al2co**) in a terminal, all options will be listed and can be used appropriately.

- B. Name directories you create in the lab system as your ROLL NO (write `mkdir <roll number>` in terminal to create directory)
- C. Edit the alignment file and remove the first line (`CLUSTAL O(1.2.4) multiple sequence alignment`).
- D. Copy alignment file from local to remote using **scp** command:
`scp <source of file> <destination>`

Example: **scp** <location to your alignment file (eg: alignment.aln)>
bt3040@10.21.49.164:~/<your folder name>/

Documentation for AL2CO is available at:

<https://github.com/TheApacheCats/al2co/blob/master/README>

2. Tabulate the topmost 10 residues with the highest and lowest conservation scores (in both Set1 and Set 2) obtained with method (i).
3. Write a program to compute the conservation score from MSA using unweighted frequency, and entropy, variance, and sum of pairs-based measures.
4. Using the program written in Q3 (unweighted frequency and entropy-based measure), compare the MSA from Clustal Omega, MAFFT, and MUSCLE. Identify the residues with (i) similar and
(ii) Different conservation scores among the three alignment methods.
5. Check the scores manually at positions 9, 11, 20, 22, and 30 (use MSA from Clustal Omega)
6. Obtain the conservation score of 1BTM, A-chain using Consurf server (<https://consurf.tau.ac.il/>)

Deadline: 10 March 2026